Solaris

Solar & wind energy forecasting for Davanagere, Karnataka. A plain-language guide to the whole project.
Written for Bhavin, 10 November 2026. Read this and you can explain every part of the project.

> **The project in one paragraph.** Solaris is a website, running on your laptop, that predicts the next 72 hours (3 days), hour by hour, of five things at Davanagere: solar power, wind power, sunlight, humidity and wind speed. It takes fresh weather forecasts from the internet every hour and runs them through a trained AI model. The big question the project answers honestly: *can machine learning beat the normal weather forecast?* Answer: yes, on all five, tested on a full year (2025) the model had never seen.

1. What it predicts (the five targets)

Target	Unit	What the unit means
Solar power	kW (kilowatt)	How much electricity the solar panels make right now. 1 kW running for 1 hour = 1 kWh = 1 "unit" on an electricity bill.
Wind power	kW	How much electricity one wind turbine makes right now.
Sunlight (GHI)	W/m² (watts per square metre)	GHI = Global Horizontal Irradiance: how strong the sunlight is on flat ground. About 1000 W/m² at a sunny noon, 0 at night.
Humidity (RH)	% (percent)	RH = Relative Humidity: how much water vapour is in the air, from 0% (dry) to 100% (saturated, like fog).
Wind speed	m/s (metres per second)	Wind speed measured 10 m above the ground. 4 m/s ≈ 14 km/h.

These five are **physically linked**: sunlight drives solar power, wind speed drives wind power, and humidity and clouds affect sunlight. That is why they are forecast together.

2. The power plant (it is imaginary, on purpose)

There is no real power plant with a meter at Davanagere that we can read. So the project uses an **imaginary plant** placed at Davanagere, with realistic, standard specifications (from `config/plant.yaml`):

- **Solar:** 1000 kWp of panels (kWp = kilowatt-peak, the panels' maximum rating; about 2,000–2,500 panels on 4–5 acres), tilted 15° facing south. An inverter (the box that converts panel power for the grid) caps output at about 800 kW.
- **Wind:** one 3,600 kW turbine, hub (centre of the blades) 80 m high. It starts turning at 3.5 m/s, reaches full power at 13 m/s, and shuts down for safety above 25 m/s.

The power numbers are labelled **"Simulated energy"** on the website. They are calculated from the weather using physics (section 4). Do not say "India produces this"; the numbers are for this one small imaginary plant.

3. Where the data comes from

Source	What it gives	Used for
NASA POWER	Hourly weather history for Davanagere from 2001 to 2025, from satellites and weather models.	The "truth" (what really happened) for training and for scoring forecasts.
Open-Meteo	Weather forecasts: past forecasts (history) and live forecasts for the next days.	Model inputs, baseline 3, and the live website every hour.
Kaggle (data-sharing website)	Real measured power from 2 solar plants in India and 1 wind turbine in Turkey.	Checking that the physics simulation is realistic.

Rules followed: the weather services are called directly over the internet (no helper libraries), and every download is saved in `data/raw/` so nothing is downloaded twice. Credits: *Weather data by Open-Meteo.com (CC BY 4.0)* and *NASA POWER data courtesy of NASA Langley Research Center*.

Time zones: all data is stored in UTC (world standard time). The website shows IST (Indian Standard Time = UTC + 5:30). Keeping one time zone inside the code avoids "5½-hour" mistakes.

4. Turning weather into power (physics)

Solar

Using **pvlib**, a standard solar-engineering library: find where the sun is in the sky each hour, work out how much sunlight lands on the tilted panels, estimate how hot the panels get (hot panels make less power), then convert to electricity with typical losses (14%) and the inverter limit.

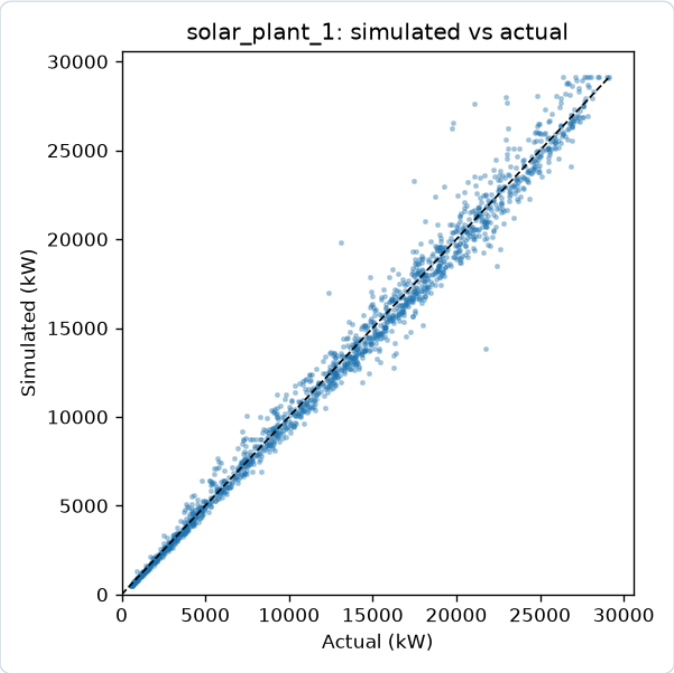
Wind

Wind is measured at 10 m but the turbine hub is at 80 m, where wind is stronger, so the speed is adjusted to hub height. Then a **power curve** (a table from the manufacturer: "at this wind speed, this many kW") gives the turbine's output.

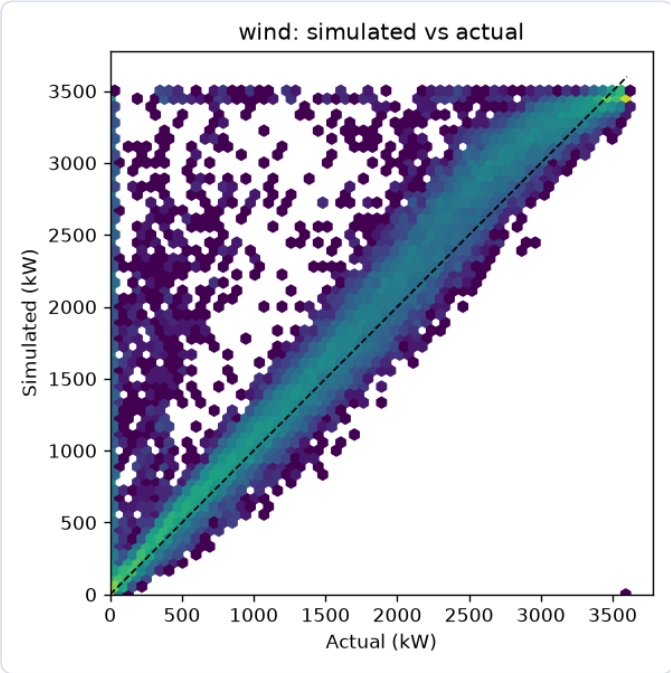
How we know the simulation is right

We ran the same physics on the weather recorded at the real Kaggle plants and compared with what those plants actually produced:

- **Solar plant 1:** error about 3% of plant size. Very good match.
- **Solar plant 2:** error about 14%. This plant's data has quality problems, so it was kept as information only and plant 1 was used as the pass/fail check.
- **Wind turbine:** passed its 14% target. Yearly energy: simulated about 12.1% of maximum vs 12.9% from the real power curve, very close.



Solar plant 1: simulated (one axis) vs real (other axis). Points on the diagonal = perfect match.



Wind turbine: simulated vs real power.

5. The models (the "contestants")

Every model gets the same job: given the past week and the weather forecast, predict the next 72 hours. Simplest to most advanced:

Model	In plain words
Persistence	"Tomorrow will be like today." The simplest possible guess.
Climatology	"Use the usual value for this month and hour." A long-term average.
Baseline 3 (the important one)	Take the professional weather forecast (Open-Meteo) and convert it to power with the physics above. This is what you would get *without* any machine learning. Every model must beat this to be worth anything.
LightGBM	Many small decision trees ("if cloud cover > 60% and hour = noon, then…") combined. Fast and strong.
LSTM / GRU	Neural networks with "memory" that read the past hours in order, like reading a sentence word by word. (LSTM = Long Short-Term Memory, GRU = Gated Recurrent Unit.)
TFT (the winner)	Temporal Fusion Transformer: a modern deep-learning model that looks at the past and the future weather forecast together and learns which inputs matter most. It forecasts all five targets at once, with a likely range.

The machine-learning models learn to **correct** the weather forecast's typical mistakes, rather than starting from nothing. Deep models (LSTM, GRU, TFT) were trained on your NVIDIA graphics card.

6. Testing honestly (why the results can be trusted)

The data is split by **time**, never shuffled:

Training 2001–2023 models learn here → **Validation** 2024 pick the best model & settings → **Test** 2025 final exam, used once

- **Why not shuffle?** Forecasting means predicting the future from the past. Shuffling would let the model peek at tomorrow while learning from yesterday, which makes scores look better than they really are.
- **Full years** in each part, so every season (summer, monsoon, winter) is included.
- **The test year was used only once** (on 9 November 2026). The code refuses to run it again, so nobody can keep trying until the numbers look good.
- Models were chosen using 2024 only. 2025 was a true surprise test.

How accuracy is measured

- **RMSE** (Root Mean Square Error): the typical size of a forecast's miss. Lower is better.
- **Skill** = how much smaller the model's RMSE is than baseline 3's. +50% means the error is cut in half. Positive = better than the weather forecast.
- **P10–P90 range:** the shaded band on the charts. It should contain the real value about 80% of the time.

7. The results

Test year 2025, TFT against baseline 3 (from `reports/evaluation/leaderboard.csv`):

Target	Baseline 3 typical miss	TFT typical miss	TFT skill
Solar power (800 kW plant)	132 kW	64 kW (about 8%)	**+51.9%**
Sunlight (GHI)	149 W/m²	72 W/m²	**+51.4%**
Humidity	11.4 % points	6.5 % points	**+42.8%**
Wind speed	1.38 m/s	1.16 m/s	**+16.2%**
Wind power (3600 kW turbine)	690 kW	552 kW (about 15%)	**+19.9%**

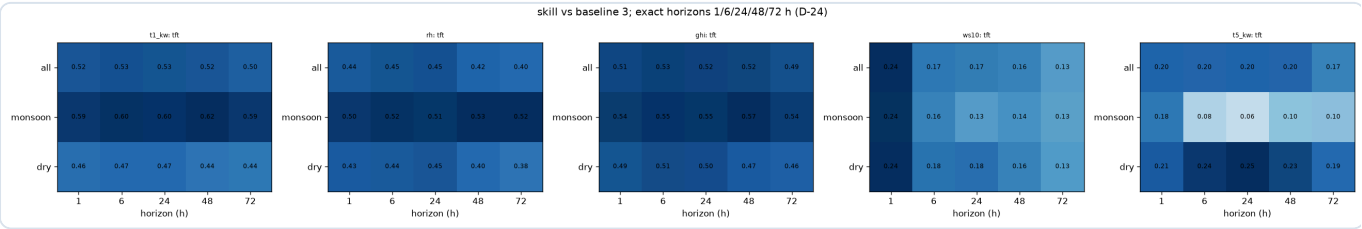
> **Main result:** TFT beats the weather-forecast method on all five targets. For solar and sunlight it cuts the error roughly in half. Wind gains least because wind is naturally jumpy and hard to predict. TFT is used on the website for all five targets.

Accuracy by day ahead (November is "dry season")

Target	Tomorrow	Day 2	Day 3
Solar power	~62 kW	~66 kW	~67 kW
Sunlight	~66 W/m²	~71 W/m²	~71 W/m²
Humidity	~7 % pts	~8 % pts	~8 % pts
Wind speed	~1.1 m/s	~1.2 m/s	~1.4 m/s
Wind power	~535 kW	~567 kW	~626 kW

Forecasts get a little less accurate the further ahead they look. These are **averages**: a clear day may be almost perfect, a sudden storm can miss more. The accuracy is the same whatever the date (11 November, 5 December, or later).

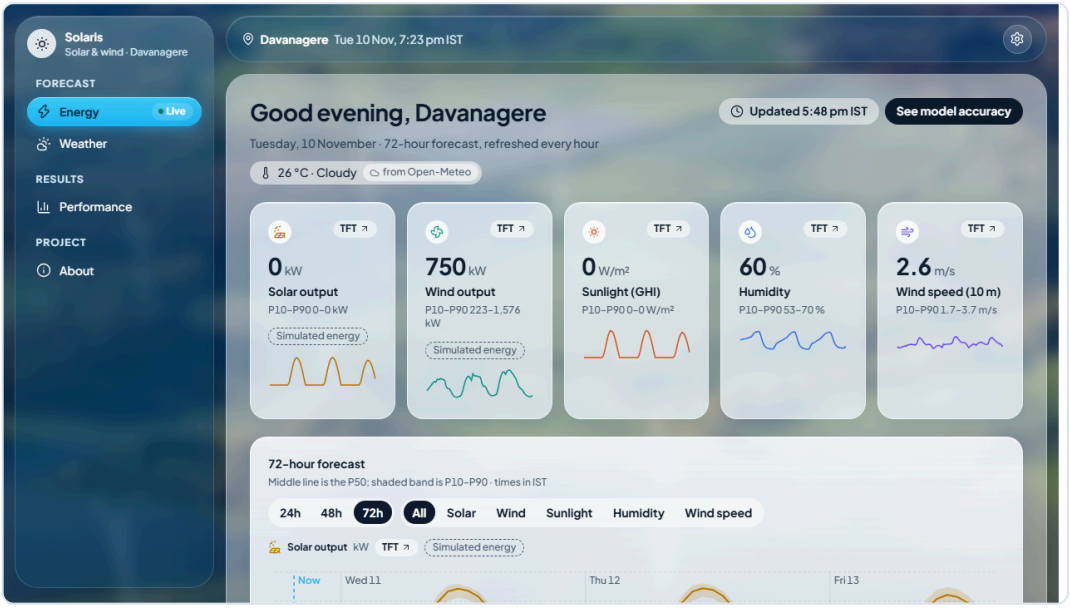
> **Honest weak spot:** the shaded P10–P90 band should catch the real value 80% of the time. On the test year it did for solar (81%) and sunlight (79%), but less for humidity (74%), wind speed (66%) and wind power (68%). So the wind bands are a bit too narrow (the model is over-confident about wind). This is stated openly as a limitation.



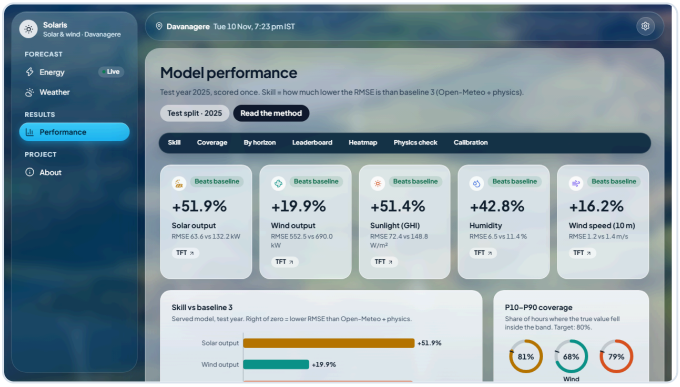
Skill heatmap (test year): where in the day and how far ahead the model beats baseline 3.

8. The website

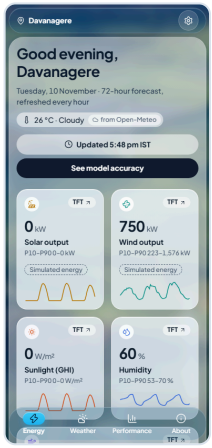
Four pages: **Energy** (solar & wind power), **Weather** (sunlight, humidity, wind), **Performance** (all the results and charts), **About** (data credits). Every hour it fetches a fresh Open-Meteo forecast, runs TFT, and saves the result in a small database (`data/app.db`). Pages load in well under half a second because they read from that saved copy.



Energy page: five cards (current value and P10–P90 range) and the 72-hour chart.



Performance page: skill, leaderboard, heatmap, coverage, physics check.



Works on a phone too.

If there is no internet: the site still opens and shows the last forecast with a yellow "may be out of date" note. That is by design.

9. How to run it

1. Connect to the internet.
2. In the project folder, run: `uv run python -m swf serve`
3. Open **http://127.0.0.1:8000** in the browser. Check the "Updated" time is recent.
4. To stop it: press **Ctrl + C** in the terminal.

Other commands (only needed to rebuild from scratch, explained in `README.md`): `data` (download and process), `train` (retrain models, needs the graphics card, hours), `evaluate` (re-score). Do **not** retrain before the viva: the saved models are the ones behind the official results.

10. The code, in the order the data flows

Folder / file	Job
`config/*.yaml`	All settings: location, plant size, model settings. Nothing is hard-coded elsewhere.
`src/swf/cli.py`	The commands: data, train, evaluate, serve.
`src/swf/fetch/`	Downloads NASA, Open-Meteo and Kaggle data; saves raw copies.
`src/swf/align/`	Cleans data and puts everything on one hourly UTC timetable.
`src/swf/physics/`	Weather → solar and wind power.
`src/swf/validate/`	Compares simulated power with the real Kaggle plants.
`src/swf/features/`	Builds the model inputs without peeking into the future; makes the time splits.
`src/swf/models/`	LightGBM, LSTM/GRU and TFT.
`src/swf/evaluate/`	Scores, leaderboard, charts, the one-time test run.
`src/swf/predict/` & `src/swf/api/`	Makes the live 72-hour forecast; the web server (FastAPI) and hourly refresh.
`frontend/`	The website (React).
`tests/`	478 automatic checks, including "no peeking into the future" tests. All pass.

Tools used: Python (main language), pandas (tables), pvlib (solar physics), LightGBM, PyTorch and darts (deep learning), FastAPI (web server), SQLite (small database), React + Vite + Tailwind + Recharts (website and charts).

11. Questions an examiner may ask (short answers)

Is the power data real?

It is simulated from real weather with standard physics, and the simulation was checked against real Kaggle plants (about 3% error on the good solar plant, wind within target).

How do you know ML beats the normal forecast?

Every model is compared with baseline 3 (the professional forecast + physics) on a year it never saw, scored once. TFT beat it on all five targets.

Why not shuffle the data?

That would let the model see the future during training and give fake-good scores. We split by time: train 2001–2023, choose on 2024, test on 2025.

Why TFT for everything?

It had the lowest error on all five targets, and using one model keeps the website and results simple and consistent.

Will the forecast be right on the day of the demo?

No forecast is exact. On average it misses solar by about 8% of plant size for tomorrow, and the shaded band shows how sure it is.

What does "truth" mean here?

The NASA POWER weather record (from satellites and weather models) plus the simulated power. There is no ground weather station at the site, which is a limitation worth mentioning.

What is a weakness?

The wind ranges are too narrow (catch the real value 66–68% instead of 80%). Also, the historical forecasts used for training are short-range, while live forecasts look 1–3 days ahead, so the model can be slightly over-confident.

Why UTC inside?

One time zone everywhere prevents 5½-hour mix-ups; IST is shown only on the website.

Can someone else reproduce it?

Yes. A fresh copy of the project was tested on 10 November using only the README instructions, and every step worked.

What next?

Put the site online for the public, and fix the wind ranges so they reach 80%.

12. Words to remember

Forecast horizon	How far ahead: 1 to 72 hours.
NWP	Numerical Weather Prediction: the computer weather forecasts that weather services make (Open-Meteo gives these).
Baseline	A simple reference method that a smarter model must beat.
Training / validation / test	Learn / choose / final exam.
Leakage	Accidentally letting a model see future information. The tests check this never happens.
P50, P10, P90	Best guess (middle), low end and high end of the likely range.
Overfitting	Memorising the training data instead of learning patterns. Prevented by validating on 2024 and stopping training when it stops improving.
API	Application Programming Interface: the address the website uses to ask the server for data (`/api/forecast`).